ITA03-MC-Assignment 3

Name: R. Thanvitha Reddy

Reg no:192425068

MOBILE COMPUTING

ASSIGNMENT 3 – CO3

Mode: Pair Work – 2 Students per Group
Duration: 1 Week
Total Marks: 50

Pair-Work Principle

Both students are responsible for understanding the complete assignment.

However, each question contains clearly identified individual responsibilities so that the contribution and understanding of each student can be independently verified.

Student A and Student B must not divide the assignment question-wise. Both students must participate in every question.

The individual responsibilities intentionally alternate across questions so that both students demonstrate evidence of:

Concept Understanding → Network Analysis → Protocol Design → Implementation/Simulation → Evaluation

Q1 – Wireless Network Architecture and Mobility Analysis 6 Marks

Part A – Introduction to Wireless Networks

Shared Group Work

Together:

- Understand the characteristics and components of wireless networks.
- Identify mobile nodes, wireless access points/base stations and communication links.
- Develop a suitable wireless network architecture.
- Identify bandwidth, connectivity and mobility constraints.
- analyse communication between mobile users and network infrastructure.
- Implement or simulate a small wireless communication scenario and test the communication flow.

Student A – Network Architecture Analysis

Student A must individually:

- Draw and represent the wireless network architecture.
- Identify the major network components and their roles.
- analyse wireless communication characteristics.
- Analyse the effect of limited bandwidth and wireless connectivity.
- analyse how mobile nodes communicate through the wireless infrastructure.

Student B – Mobility and Performance Analysis Student

B must individually:

- Design representative mobility scenarios.
- Analyse handoff and connectivity interruption.
- Study packet loss and transmission delay.
- Analyse the effect of changing network conditions.
- Compare communication behaviour under different mobility conditions.

Part B – Wireless Mobility Scenario

Shared Group Work

Together, develop and analyse a scenario in which a mobile user moves between wireless coverage areas.

Student A – Analytical Investigation

Student A must:

- Define successful communication and interrupted communication cases.
- Identify the sequence of mobility events.
- Record communication delay and packet delivery behaviour.
- Investigate increasing mobility conditions.
- Present the analytical evidence.

Student B – Mobility Logic

Student B must:

- Formulate the mobility decision sequence.
- analyse when a mobile node changes its point of attachment.
- Identify which communication conditions are affected.
- analyse the termination or completion condition for the scenario.
- Relate repeated mobility to communication efficiency.

Individual Verification

Faculty may ask Student A to analyse the wireless architecture and Student B to analyse the mobility scenario. Therefore, primary responsibility does not mean exclusive knowledge.

Q2 – Mobile IP and Mobility Management **7 Marks**

Shared Group Work

The pair must:

- Study Mobile IP architecture and terminology.
- Identify Mobile Node, Home Agent, Foreign Agent and Correspondent Node.
- analyse agent discovery and registration.
- Analyse packet delivery to a mobile node.
- analyse tunnelling and mobility support.
- Conduct a small simulation or worked example of Mobile IP communication.

Student A – Mobile IP Operation

Student A is individually responsible for:

- analyseing the role of the Mobile Node.
- analyseing the role of the Home Agent.
- analyseing the role of the Foreign Agent.
- analyseing the role of the Correspondent Node.
- Describing registration and location updating.
- analyseing the packet-delivery and tunnelling process.

Student B – Mobile IP Challenge Analysis Student

B is individually responsible for:

- Identifying Mobile IP challenges.
- Analysing registration delay.
- Analysing tunnelling overhead.
- analyseing triangular routing.
- Analysing packet loss during mobility.
- interpreting security and scalability considerations.

Joint Design Question

Together answer:

If a mobile user frequently changes its network of attachment during an active communication session, what problems can occur in Mobile IP and how can they be reduced?

Evidence Expected

- Mobile IP architecture diagram.
- Registration/packet-flow sequence.
- Identification of mobility-related delays and overhead.
- Reasoned justification of the proposed improvement.

Cross-Understanding Requirement

Student A must be able to analyse the major Mobile IP challenges.

Student B must be able to analyse the complete Mobile IP registration and packet-delivery process.

Both students must be able to defend the answer.

Q3 – Cellular Networks versus Ad Hoc Wireless Networks 7 Marks

Shared Group Work

Together:

- Study cellular wireless network architecture.
- Study ad hoc wireless network architecture.
- Identify infrastructure-based and infrastructure-less communication.
- Compare mobility and topology characteristics.
- Compare routing requirements.
- Develop a technical comparison diagram and table.

Student A – Cellular Network Investigation Student

A must individually:

- Represent cellular network architecture.
- analyse the role of base stations and infrastructure.
- Analyse centralized communication.

- analyse handoff between cells.
- Analyse scalability and coverage.
- Identify suitable cellular applications.

Student B – Ad Hoc Network Investigation Student

B must individually:

- Represent ad hoc network architecture.
- analyse communication between mobile nodes.
- Analyse dynamic topology.
- analyse multi-hop communication.
- Analyse the absence of fixed infrastructure. ● Identify suitable ad hoc applications.

Comparative Analysis

Parameter	**Cellular Network**	**Ad Hoc Network**
Infrastructure		
Communication		
Routing		
Mobility		
Topology		
Scalability		
Handoff		
Failure Handling		
Applications		

Joint Engineering Decision Together

determine:

For an emergency/disaster environment where fixed infrastructure may be unavailable, which network architecture would you select and why?

Cross-Defence

Student A must analyse why cellular networks depend on infrastructure.

Student B must analyse how ad hoc networks continue communication without fixed infrastructure.

Q4 – Applications of Mobile IP and Routing Requirements 7 Marks

Shared Group Work

Together:

- Identify real-world applications requiring mobility support.
- Select one application scenario such as emergency communication, mobile healthcare or intelligent transportation.
- Identify the mobile entities and communication requirements.
- Apply Mobile IP concepts to the selected scenario.
- Identify the routing requirements created by mobility.
- Evaluate the suitability of the proposed solution.

Student B – Application Architecture

Student B must individually:

- Select and describe an appropriate Mobile IP application.
- Identify mobile nodes and communication endpoints.
- Design the application network architecture.
- analyse how Mobile IP supports user mobility.
- Identify required network components.
- analyse the communication flow.

Student A – Performance and Requirement Analysis

Student A must individually:

- Analyse communication delay.
- Analyse packet delivery and connectivity.
- Identify mobility requirements.
- Analyse interruption during movement.
- Identify security and reliability requirements. ● Evaluate Mobile IP suitability.

Q5 – Proactive Routing Protocols in Mobile Networks 6 Marks Shared Group Work

Together:

- Study proactive routing protocols.
- Understand continuous routing-table maintenance.
- Select a suitable protocol such as DSDV.
- Develop a mobile-node topology.
- Analyse routing-table updates.
- Evaluate routing behaviour under mobility.

Student A – Protocol Design

Student A must individually:

- analyse the principle of proactive routing.
- analyse DSDV operation.
- analyse routing-table maintenance.
- analyse sequence numbers and route updates.
- Draw a sample routing topology.
- Demonstrate route information before and after a topology change.

Student B – Performance Investigation Student

B must individually:

- Analyse routing-control overhead.
- Analyse route availability.
- Study convergence behaviour.
- Analyse control-message transmission.
- Compare behaviour for different network sizes. ● Evaluate delay and packet delivery.

Q6 – Reactive Routing and Protocol Comparison 7 Marks

Student A – Reactive Routing Investigation

Student A takes primary responsibility for reactive routing.

- analyse the concept of reactive routing.
- Study AODV.
- analyse Route Request (RREQ).

- analyse Route Reply (RREP). ● analyse Route Error (RERR).

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-

  Analyse route discovery and route maintenance. analyse the effect of mobility on route discovery.

Student B – Comparative Verification

Student B must provide a comparative analysis of AODV and DSDV.

- Compare route discovery mechanisms.
- Compare routing-table requirements.
- Analyse control overhead.
- Compare initial communication delay.
- Analyse mobility handling.
- Identify suitable application scenarios.
- Justify the selection of a protocol for a given network condition.

Joint Question

For a highly dynamic mobile network with frequent topology changes, would you select DSDV or AODV? Justify your decision using routing overhead, delay and mobility considerations.

DAY-4 DESIGN CHALLENGE

Q7 – Design a Mobile Emergency Communication Network 10 Marks

The Day-4 problem must be solved collaboratively.

Neither student may work on this question independently and simply combine the answers later.

Scenario

During a natural disaster, ambulances, police vehicles, rescue teams and medical personnel move continuously across different geographical areas. Fixed communication infrastructure may become unavailable, while mobile users must exchange critical information with minimum delay.

Stage 1 – Independent Design Proposal

Before interpreting the solution with each other:

-
-

Student A

Independently propose:

A suitable wireless network architecture.
A mobility-management strategy.

- The role of Mobile IP.
- A possible communication model.

Student B

Independently propose:

- A suitable routing strategy.
- Proactive/reactive protocol selection.
- Route maintenance approach.
- Failure-handling mechanism. ● Performance parameters.

Each student's initial proposal must be retained in the submission.

Stage 2 – Compare the Two Designs

The pair must compare where the designs agree, where they differ, strengths and weaknesses, Mobile IP requirements, routing suitability, expected overhead, expected delay and handling of node mobility.

Stage 3 – Joint Final Design

Using the strongest ideas from the interpretion, jointly develop the final mobile emergency communication solution.

- Network architecture showing mobile nodes, wireless links and routing infrastructure.
- Mobile IP components: Mobile Node, Home Agent, Foreign Agent and Correspondent Node.
- Registration and tunnelling process.
- Selected routing protocol and justification.
- Failure handling for node/link failure and network disconnection.
- Performance evaluation using delay, packet delivery and routing overhead.

Stage 4 – Divide the Validation Responsibilities

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Student A – Analytical Validation

- Analyse Mobile IP overhead.
- Analyse mobility-related delay. ● Evaluate packet delivery.

  Analyse communication interruption.
  Justify the selected mobility-management mechanism.

Student B – Experimental Validation

- Design network test cases.
- Test different numbers of mobile nodes.
- Test different mobility conditions.
- Compare proactive and reactive routing behaviour.
- Record delay and packet-delivery results.
- Present experimental evidence.

Stage 5 – Cross-Defence

Student A must analyse: Why was the selected routing protocol appropriate for the emergency mobile network?

Student B must analyse: How does Mobile IP maintain communication when a mobile node changes its network location?

Stage 6 – Joint Evaluation

- Why the proposed network architecture is suitable.
- Why Mobile IP is required or not required.
- Why the selected routing protocol is appropriate.
- What network overhead is introduced.
- How mobility affects performance.
- What happens when connectivity is lost.
- Under what conditions the proposed solution may not be appropriate.

Conclude: What characteristics should a network designer analyse before selecting Mobile IP and a routing protocol for a highly mobile wireless environment?

CONTRIBUTION RECORD

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Question	**Student A – Primary Contribution**	**Student B – Primary Contribution**	**Joint Work**
Q1	Network Architecture	Mobility Analysis	Wireless Network Analysis
Q2	Mobile IP Operation	Mobile IP Challenges	Mobile IP Architecture

Q3	Cellular Network	Ad Hoc Network	Comparative Analysis
Q4	Performance Analysis	Application Architecture	Application Evaluation
Q5	Protocol Design	Performance Investigation	Proactive Routing
Q6	Reactive Routing	DSDV–AODV Comparison	Routing Evaluation
Q7	Analytical Validation	Experimental Validation	Final Network Design

INDIVIDUAL VERIFICATION

Faculty may select any question, including a component for which the student was not designated as the primary contributor.

- analyse wireless network architecture.
- analyse Mobile IP operation and registration.
- analyse mobility challenges.
- Compare cellular and ad hoc wireless networks.
- analyse applications of Mobile IP.
- analyse proactive routing.
- analyse reactive routing.
- Compare DSDV and AODV.
- Justify protocol selection.
- Interpret experimental results.
- Justify network design decisions.
- Respond to a modified network condition.

A student should not answer: "My partner did this section."

Understanding of the complete submitted work is a requirement of the assignment.

NOTE: RECOMMENDED FACULTY VERIFICATION MODEL (VIVA)

For each student, conduct a brief 4–5 minute verification.

Question 1 – Primary responsibility

Ask one technical question from the component for which the student had primary responsibility.

Question 2 – Partner's responsibility

Ask one question from a component primarily completed by the partner.

Question 3 – Transfer question

Modify a condition and ask the student to reason about the consequence.

- If a mobile node moves from one network to another during an active session, what happens to its IP connectivity and how does Mobile IP address the problem?
- If the number of mobile nodes increases significantly, how would AODV and DSDV differ in routing overhead?
- If the network topology changes very frequently, which routing approach would you select and why?
- If the fixed infrastructure becomes unavailable during a disaster, how would an ad hoc network support communication?

This verification establishes individual attainment without requiring a second complete individual assignment.

SIMATS ENGINEERING

MOBILE COMPUTING

STANDARD FOR ALGORITHMIC PSEUDOCODE

Applicable to: Mobile COmputing Assignments and Exercises

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1. What is Mobile Computing Pseudocode?

Mobile Computing pseudocode is a clear, precise and language-independent representation of a mobile networking algorithm. It describes mobility management, packet forwarding, route discovery, route maintenance, handoff, and routing decisions without depending on Java, Python, C, C++, Android, or simulator-specific syntax.

A good Mobile Computing algorithm must make the network state and communication logic visible. It should show what happens when a mobile node moves, when a route

exists or fails, when a registration is required, and when a packet must be forwarded through a tunnel or discovered route.

Remember

- Pseudocode is not English prose.
- Pseudocode is not programming code.
- Pseudocode is an implementation-independent specification of mobile-network logic.
- Every important networking decision must be explicit.
- Abstraction is permitted; ambiguity is not.

2. Standard Structure

Use the following structure for a complete Mobile Computing algorithm.

ALGORITHM AlgorithmName(parameters)

INPUT:

Clearly describe nodes, packets, addresses, routes or network state

OUTPUT:

Clearly describe the expected mobility/routing result

BEGIN

INITIALISE required variables/data structures

Detect or receive relevant network information

Execute mobility/routing decisions

UPDATE network state

RETURN result

END

Example

ALGORITHM CheckMobileNodeLocation(MN, HomeNetwork, CurrentNetwork)

INPUT:

MN – mobile node

HomeNetwork – node's registered home network CurrentNetwork

– network where MN is currently attached

OUTPUT:

Mobility status

BEGIN

IF CurrentNetwork = HomeNetwork THEN

status ← HOME_NETWORK ELSE status

← VISITED_NETWORK

END IF

RETURN status

END

3. Standard Keywords and Notation

PURPOSE	**USE**
Declare algorithm	ALGORITHM
Specify input	INPUT:
Specify output	OUTPUT:
Start algorithm	BEGIN
End algorithm	END
Assignment	←
Condition	IF ... THEN
Alternative	ELSE
Multiple conditions	ELSE IF
Loop	FOR ... DO / WHILE ... DO
Return result	RETURN

Boolean operations	AND, OR, NOT
Packet operation	FORWARD / DELIVER / DROP
Route operation	ADD ROUTE / REMOVE ROUTE / UPDATE ROUTE
Message operation	SEND / RECEIVE
Timer	START TIMER / EXPIRE TIMER

Use explicit state names such as HomeAddress, CareOfAddress, NextHop, SequenceNumber, HopCount, RouteState, LinkState and RegistrationStatus.

4. Wireless Network Representation

Represent a mobile network as a graph G = (V, E), where V is the set of mobile nodes and infrastructure nodes, and E represents currently usable communication links.

NETWORK G = (V, E)

V = {mobile nodes, base stations, access points, gateways}

E = {wireless links currently available} link(u,v) =

communication link between u and v

cost(u,v) = communication cost of link (u,v) status(u,v)

∈ {UP, DOWN}

Network State

STATE:

Location[MN] = current network/location of mobile node MN

NextHop[D] = next hop toward destination D

RouteState[D] ∈ {VALID, INVALID, UNKNOWN}

SequenceNumber[D] = latest known destination sequence number

HopCount[D] = number of hops in current route

LastUpdate[D] = time of most recent route information

5. Wireless Network Discovery

Wireless nodes must determine whether they are connected to a suitable network and whether neighboring nodes are reachable. The discovery procedure is especially important in dynamic environments.

ALGORITHM DiscoverWirelessNeighbors(MN, TransmissionRange)

INPUT:

MN – mobile node

TransmissionRange – maximum discovery range

OUTPUT:

NeighborSet

BEGIN

NeighborSet ← EMPTY_SET

BROADCAST HELLO from MN

WAIT for neighbor responses

FOR each Response received DO

IF SignalStrength(Response) > MIN_SIGNAL THEN

IF Distance(MN, Response.Node) ≤ TransmissionRange THEN

ADD Response.Node to NeighborSet

END IF

END IF

END FOR

RETURN NeighborSet

END

6. Mobile IP – Basic Operation

Mobile IP allows a mobile node to retain a permanent home address while changing its point of network attachment. The algorithm must maintain a binding between the permanent home address and the current care-of address.

Mobile IP Entities

- Mobile Node (MN)
- Home Network
- Home Agent (HA)
- Foreign Network / Visited Network
- Foreign Agent (FA) in traditional Mobile IPv4 terminology
- Care-of Address (CoA)
- Correspondent Node (CN)

ALGORITHM MobileIPLocationUpdate(MN, HA, CurrentNetwork)

INPUT:

MN – mobile node

HA – home agent

CurrentNetwork – visited network

OUTPUT:

Registered care-of address

BEGIN

IF CurrentNetwork = MN.HomeNetwork THEN

CoA ← MN.HomeAddress

RegistrationStatus ← HOME

ELSE

CoA ← ObtainCareOfAddress(CurrentNetwork)

SEND RegistrationRequest(MN.HomeAddress, CoA) TO HA

WAIT FOR RegistrationReply

IF RegistrationReply = ACCEPTED THEN

RegistrationStatus ← REGISTERED

ELSE

RegistrationStatus ← FAILED

END IF

END IF

RETURN CoA, RegistrationStatus

END

7. Mobile IP Agent Discovery ALGORITHM

MobileIPAgentDiscovery(MN)

INPUT: MN –

mobile node

OUTPUT:

Detected mobility agent or network information

BEGIN

LISTEN for AgentAdvertisement

IF Advertisement received THEN

Extract NetworkAddress

Extract AgentType

IF AgentType = HOME_AGENT THEN

Set HomeAgentAddress

ELSE IF AgentType = FOREIGN_AGENT THEN

Set ForeignAgentAddress

END IF

RETURN AgentType, NetworkAddress

ELSE

SEND AgentSolicitation

WAIT for Advertisement

IF Advertisement received THEN

RETURN detected agent information

ELSE

RETURN AGENT_NOT_FOUND

END IF

END IF

END

8. Mobile IP Registration

ALGORITHM RegisterMobileNode(MN, HA, CoA)

INPUT:

MN – mobile node

HA – home agent CoA –

current care-of address

OUTPUT:

Registration result

BEGIN

Create RegistrationRequest

Request.HomeAddress ← MN.HomeAddress

Request.CareOfAddress ← CoA

Request.Lifetime ← RequestedLifetime

Request.Identifier ← GenerateRegistrationIdentifier()

Authenticate Request

SEND Request TO HA

WAIT FOR RegistrationReply

IF Reply.Status = ACCEPTED THEN

Binding[MN.HomeAddress] ← CoA

RegistrationExpiry ← CurrentTime + Reply.Lifetime

RETURN REGISTRATION_SUCCESS

ELSE

RETURN REGISTRATION_FAILURE

END IF

END

9. Mobile IP Packet Tunnelling

The home agent can forward packets toward the mobile node's current location by tunnelling them to the registered care-of address.

ALGORITHM TunnelPacketToMobileNode(Packet, DestinationHomeAddress)

INPUT:

Packet – original IP packet

DestinationHomeAddress – mobile node home address

OUTPUT:

Packet delivered through tunnel or DROP

BEGIN

IF Binding[DestinationHomeAddress] does not exist THEN

DROP Packet

RETURN DELIVERY_FAILURE

END IF

CoA ← Binding[DestinationHomeAddress]

EncapsulatedPacket ← ENCAPSULATE(Packet, CoA)

FORWARD EncapsulatedPacket TO CoA

IF TunnelDeliverySuccessful THEN

Decapsulate at tunnel endpoint

DELIVER original Packet TO MobileNode

RETURN DELIVERY_SUCCESS

ELSE

RETURN DELIVERY_FAILURE

END IF

END

10. Mobile IP Challenges

CHALLENGE	**ALGORITHMIC IMPLICATION**
Handoff delay	Detect movement and update location quickly
Registration overhead	Limit unnecessary signalling
Packet loss	Buffer, retransmit or reroute when appropriate
Triangular routing	Evaluate indirect path and forwarding delay
Security	Authenticate registration and protect mobility messages
Scalability	Avoid excessive control traffic
Wireless errors	Detect link failure and retry or select alternate path
Changing bandwidth	Adapt routing and transmission decisions

Challenge Detection Algorithm

ALGORITHM DetectMobileIPChallenge(MN, Route, Registration)

INPUT:

MN, current route, registration state

OUTPUT:

Identified challenge

BEGIN

IF RegistrationStatus = FAILED THEN

RETURN REGISTRATION_PROBLEM

ELSE IF HandoffInProgress = TRUE THEN

RETURN HANDOFF_DELAY

ELSE IF PacketLossRate > MAX_LOSS THEN

RETURN PACKET_LOSS

ELSE IF RouteDelay > MAX_DELAY THEN RETURN

ROUTING_DELAY

ELSE IF SecurityValidation = FAILED THEN

RETURN SECURITY_PROBLEM

ELSE

RETURN NO_MAJOR_PROBLEM

END IF

END

11. Handoff Management

ALGORITHM PerformHandoff(MN, OldNetwork, NewNetwork)

INPUT:

MN – mobile node

OldNetwork – previous point of attachment

NewNetwork – target network

OUTPUT:

Handoff status

BEGIN

Detect NewNetwork

Measure signal quality

IF SignalQuality(NewNetwork) < MIN_SIGNAL THEN

RETURN HANDOFF_REJECTED

END IF

Obtain new network address or care-of address

Pause non-critical transmissions if required

UPDATE mobility registration

UPDATE routing information

IF RegistrationStatus = SUCCESS THEN

SWITCH active attachment TO NewNetwork

Resume data transmission

RETURN HANDOFF_SUCCESS

ELSE

Restore OldNetwork if available

RETURN HANDOFF_FAILURE

END IF

END

12. Cellular vs Ad Hoc Wireless Networks

PARAMETER	**CELLULAR NETWORK**	**AD HOC NETWORK**
Infrastructure	Base stations / access infrastructure	No fixed infrastructure required
Topology	Planned cells	Dynamic topology
Routing	Infrastructure-assisted	Distributed multi-hop routing

Node role	Mobile device mainly communicates through infrastructure	A node can act as host and router
Mobility support	Cellular mobility management	Distributed route discovery/maintenance
Deployment	Requires pre-existing infrastructure	Can be rapidly deployed
Typical environment	Public cellular service	Emergency, military, temporary or field networks
Failure behaviour	Infrastructure failure can affect service	Alternate peer routes may be possible

13. Selecting Cellular or Ad Hoc Architecture ALGORITHM

SelectWirelessArchitecture(Scenario)

INPUT:

InfrastructureAvailability

NodeMobility

DeploymentTime

CoverageRequirement

MultiHopRequirement

OUTPUT:

Recommended architecture

BEGIN

IF InfrastructureAvailability = AVAILABLE AND

CoverageRequirement = LARGE THEN

Architecture ← CELLULAR

ELSE IF DeploymentTime = IMMEDIATE AND

InfrastructureAvailability = UNAVAILABLE THEN

Architecture ← AD_HOC

ELSE IF MultiHopRequirement = REQUIRED THEN

Architecture ← AD_HOC

ELSE

Architecture ← CELLULAR

END IF

RETURN Architecture

END

14. Applications of Mobile IP

- Mobile employees accessing organizational services while moving between networks.
- Continuous Internet access across different points of attachment.
- Emergency response and disaster recovery communication.
- Mobile healthcare and ambulance communication.
- Transportation and fleet monitoring.
- Field engineering and remote maintenance.
- Temporary communication networks.
- Location-independent access to network applications. ALGORITHM

  SelectMobileIPApplication(Requirement)

INPUT:

Requirement – mobility and communication requirement

OUTPUT:

Suitable application category

BEGIN

IF Requirement requires continuous Internet access while moving THEN

RETURN MOBILE_INTERNET_ACCESS

ELSE IF Requirement involves emergency vehicle mobility THEN

RETURN EMERGENCY_COMMUNICATION

ELSE IF Requirement requires remote patient/device connectivity THEN

RETURN MOBILE_HEALTHCARE

ELSE IF Requirement involves moving vehicles or fleets THEN

RETURN TRANSPORTATION

ELSE

RETURN GENERAL_MOBILE_ACCESS

END IF

END

15. Routing Protocols in Mobile Networks

Routing protocols determine how packets travel through changing mobile topologies. For this unit, routing is studied using the two major approaches: proactive and reactive.

Routing State

STATE:

RouteTable[Destination] = {NextHop, HopCount, SequenceNumber, State}

State ∈ {VALID, INVALID, UNKNOWN}

NeighborTable = currently reachable neighbors

RouteRequestID = identifier for route discovery

RouteLifetime = validity period

16. Proactive Routing

A proactive protocol maintains route information before a packet transmission request occurs. This can reduce route-discovery delay but may create periodic control overhead.

ALGORITHM ProactiveRouteMaintenance(Node)

INPUT:

Node – mobile node OUTPUT:

Updated routing table

BEGIN

INITIALISE RouteTable

PERIODICALLY DO

Broadcast current routing information

RECEIVE routing updates from neighbors

FOR each Update DO

Validate Update

FOR each Destination in Update DO

Calculate NewMetric

IF NewMetric < CurrentMetric(Destination) OR

Update.SequenceNumber > CurrentSequenceNumber(Destination) THEN

UPDATE RouteTable[Destination]

Set NextHop and HopCount

END IF

END FOR

END FOR

Remove expired or invalid entries

END PERIODICALLY

RETURN RouteTable

END

17. Proactive Routing – DSDV-Style Sequence-Number Update

A destination-sequenced distance-vector approach uses route metrics and sequence numbers to maintain fresh routes and reduce routing loops. ALGORITHM UpdateDSDVRoute(Node, ReceivedRoute)

INPUT:

Node – receiving node

ReceivedRoute = {Destination, NextHop, HopCount, SequenceNumber}

OUTPUT:

Updated routing table

BEGIN

Destination ← ReceivedRoute.Destination

NewHopCount ← ReceivedRoute.HopCount + 1

NewSequence ← ReceivedRoute.SequenceNumber

IF Destination not in RouteTable THEN

ADD ReceivedRoute with NextHop = Node.Neighbor

ELSE IF NewSequence > RouteTable[Destination].SequenceNumber THEN

REPLACE RouteTable[Destination]

ELSE IF NewSequence = RouteTable[Destination].SequenceNumber AND

NewHopCount < RouteTable[Destination].HopCount THEN

REPLACE RouteTable[Destination]

END IF

RETURN RouteTable END

18. Reactive Routing

A reactive protocol discovers a route only when a source needs to send data and no valid route is available. Route discovery and route maintenance are therefore demand-driven.

ALGORITHM ReactiveRouteDiscovery(Source, Destination)

INPUT:

Source – requesting node Destination

– required destination

OUTPUT:

Route or ROUTE_NOT_FOUND

BEGIN

IF RouteTable[Destination].State = VALID THEN

RETURN RouteTable[Destination]

END IF

RouteRequestID ← RouteRequestID + 1

Create RouteRequest(Source, Destination, RouteRequestID)

Broadcast RouteRequest to neighbors

WAIT for RouteReply or RouteFailure

IF RouteReply received THEN

Establish reverse path

Establish forward path

Mark RouteTable[Destination] as VALID

RETURN RouteTable[Destination]

ELSE

RETURN ROUTE_NOT_FOUND

END IF

END

19. Reactive Routing – AODV-Style Route Request

ALGORITHM ProcessRouteRequest(Node, Request)

INPUT:

Node – current receiving node

Request = {Source, Destination, RequestID, HopCount}

OUTPUT:

Route discovery action

BEGIN

IF Request already processed THEN

DISCARD Request

RETURN

END IF

Record reverse route toward Request.Source

Mark Request as processed

IF Node = Request.Destination THEN

Create RouteReply

SEND RouteReply toward Source

RETURN

END IF

IF ValidRouteTo(Request.Destination) THEN

Create RouteReply using valid route

SEND RouteReply toward Source

ELSE

Request.HopCount ← Request.HopCount + 1

FOR each Neighbor in NeighborTable DO

FORWARD Request TO Neighbor

END FOR

END IF

END

20. Reactive Route Maintenance

ALGORITHM MaintainReactiveRoute(Source, Destination)

INPUT: Source and

Destination

OUTPUT:

Route status

BEGIN

MONITOR active route

IF LinkFailureDetected THEN

Mark failed link as INVALID

IF AlternateValidRoute exists THEN

Select alternate route

RETURN ALTERNATE_ROUTE_USED

ELSE

Generate RouteError

SEND RouteError toward affected source

IF Source still requires communication THEN

Initiate new RouteDiscovery

ELSE

RETURN ROUTE_TERMINATED

END IF

END IF

END IF

RETURN ROUTE_ACTIVE

END

21. Proactive vs Reactive Routing

FEATURE	PROACTIVE	REACTIVE
Route availability	Maintained before data transmission	Obtained when required
Route discovery delay	Low	Present when route is unknown
Control overhead	Periodic/continuous	Demand-driven
Stale information	Possible	Route is generally discovered when needed
Frequent traffic	Often suitable	May repeatedly use discovered routes
Infrequent traffic	May waste control traffic	Often efficient
Topology changes	Routing tables updated periodically	Route repair/discovery triggered
Typical example	DSDV-style	AODV-style

22. Proactive or Reactive – Decision Algorithm

ALGORITHM SelectRoutingApproach(TrafficRate, MobilityRate, DelayRequirement, Bandwidth)

INPUT:

TrafficRate – frequency of data communication

MobilityRate – frequency of topology changes

DelayRequirement – acceptable route setup delay

Bandwidth – available bandwidth OUTPUT:

Recommended routing approach

BEGIN

IF DelayRequirement = VERY_LOW AND TrafficRate = HIGH THEN

Recommendation ← PROACTIVE

ELSE IF TrafficRate = LOW AND Bandwidth = LIMITED THEN

Recommendation ← REACTIVE

ELSE IF MobilityRate = HIGH AND TrafficRate = INTERMITTENT THEN

Recommendation ← REACTIVE

ELSE IF TrafficRate = HIGH AND DelayRequirement = LOW THEN

Recommendation ← PROACTIVE

ELSE

Recommendation ← EVALUATE_BOTH

END IF

RETURN Recommendation

END

23. Route Selection and Metric Calculation

FUNCTION CalculateRouteMetric(HopCount, Delay, LossRate, EnergyCost)

INPUT:

HopCount, Delay, LossRate, EnergyCost

OUTPUT:

RouteMetric

BEGIN

RouteMetric ←

W1 × HopCount +

W2 × Delay +

W3 × LossRate +

W4 × EnergyCost

RETURN RouteMetric

END

ALGORITHM SelectBestRoute(RouteSet)

INPUT:

RouteSet – candidate routes

OUTPUT:

BestRoute

BEGIN

BestRoute ← NULL

BestMetric ← INFINITY

FOR each Route in RouteSet DO

Metric ← CalculateRouteMetric(Route.HopCount,

Route.Delay,

Route.LossRate,

Route.EnergyCost)

IF Metric < BestMetric THEN

BestMetric ← Metric

BestRoute ← Route

END IF

END FOR

RETURN BestRoute

END

24. Packet Forwarding in a Mobile Network ALGORITHM

ForwardMobilePacket(Node, Packet)

INPUT:

Node – current forwarding node

Packet – data packet

OUTPUT:

Delivery, forwarding or drop decision

BEGIN

Destination ← Packet.Destination

IF Node = Destination THEN

DELIVER Packet

RETURN DELIVERED

END IF

IF ValidRouteTo(Destination) THEN

NextHop ← RouteTable[Destination].NextHop

IF LinkStatus(NextHop) = UP THEN

FORWARD Packet TO NextHop

RETURN FORWARDED

ELSE

Mark route as INVALID

RETURN ROUTE_FAILURE

END IF

ELSE

RETURN ROUTE_DISCOVERY_REQUIRED

END IF

END

25. Routing Loop Detection ALGORITHM

DetectRoutingLoop(Path) INPUT:

Path – ordered sequence of nodes

OUTPUT:

TRUE if loop exists, otherwise FALSE

BEGIN

Visited ← EMPTY_SET

FOR each Node in Path DO

IF Node IN Visited THEN

RETURN TRUE

END IF

ADD Node TO Visited

END FOR

RETURN FALSE

END

26. Manual Trace / Desk Check

Mobile routing pseudocode must be manually traceable. Record the node, current route state, received message, decision, next hop, and resulting route table state.

STEP	**NODE**	**EVENT**	**DECISION**	**ROUTE STATE**
1	S	Needs D	No valid route	UNKNOWN
2	S	Broadcast RREQ	Forward to neighbors	DISCOVERY
3	A	Receives RREQ	Record reverse route	DISCOVERY

4	D	Receives RREQ	Generate RREP	VALID
5	S	Receives RREP	Install route S→A→D	VALID
6	S	Data packet	Forward to A	ACTIVE

27. What Will NOT Be Accepted as Mobile Routing Pseudocode?

Too vague:

Start

Find route

Send packet

Handle movement

Display result

Stop

There is insufficient computational logic.

Programming code disguised as pseudocode:

for(int i=0; i<n; i++) { if(route[i].valid)

send(packet);

}

Acceptable algorithmic form:

FOR each Route in RouteTable DO

IF Route.State = VALID THEN

FORWARD Packet TO Route.NextHop

RETURN FORWARDED

END IF

END FOR

RETURN ROUTE_DISCOVERY_REQUIRED

28. Programming-Language Syntax Should Not Be Used

- Avoid Java, Python, C, C++, Android API calls, or simulator-specific commands unless actual implementation code is explicitly requested.
- Use SEND, RECEIVE, FORWARD, DELIVER, DROP, UPDATE, ADD ROUTE and REMOVE ROUTE.
- Use ← for assignment.
- Use explicit network conditions instead of vague statements such as 'handle mobility'.

29. Comments

// Detect whether the current link is still usable

IF LinkStatus(NextHop) = DOWN THEN

Mark Route INVALID

Initiate route repair

END IF

Comments may clarify a non-obvious networking decision, but comments cannot replace the routing logic. The supplied DAA standard similarly requires the pseudocode itself to analyse how the result is obtained.
□filecite□turn0file0□L271-L277□

30. Quality Checklist

- Inputs are clearly defined.
- Outputs are clearly defined.
- Mobile-network states have clear meanings.
- Addresses, routes and sequence information are initialised.
- Packet forwarding conditions are explicit.
- Registration and handoff conditions are explicit.
- Route discovery and route repair have clear termination.
- Link failure and unavailable-route cases are addressed.
- The algorithm can be manually traced.

- Programming-language-specific syntax is avoided.
- Another programmer could implement the algorithm without guessing the logic.

31. How Mobile Computing Pseudocode Will Be Assessed

CRITERION	WHAT IS EXPECTED
Algorithmic Correctness	The mobility or routing logic correctly solves the stated problem.
Completeness	Inputs, outputs, state, decisions, loops, packet actions and termination are sufficiently specified.
Precision	Conditions such as route validity, link state, registration status and sequence number are explicit.
Networking Understanding	The solution correctly reflects Mobile IP, cellular/ad hoc networking or routing concepts.
Appropriate Abstraction	Network operations are represented clearly without unnecessary programming syntax.
Traceability	A small network scenario can be manually executed using the pseudocode.
Technical Justification	The selected proactive/reactive or mobility mechanism is supported by network characteristics.

32. Final Test

Before submitting any Mobile Computing pseudocode, ask:

- Can another competent programmer implement my algorithm without asking what I meant?
- Can the same algorithm be implemented in Java, Python, C/C++ or a network simulator without changing its underlying logic?
- Does the algorithm explicitly handle mobility, routing state and failure conditions?

- Can I trace the algorithm using a small mobile-network example?

33. THE MOBILE COMPUTING DESIGN RULE

UNDERSTAND → MODEL NETWORK → DEFINE STATE → DESIGN ALGORITHM → WRITE PSEUDOCODE → TRACE → TEST → REFINE → IMPLEMENT

Do not write the program first and convert it into pseudocode afterwards. Design the mobile-network behaviour first. The program or simulator is the implementation of that design. This follows the source handbook's final design principle:

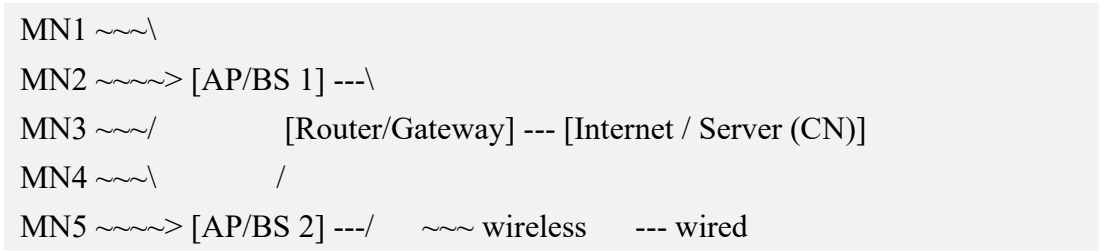
THINK → DESIGN → WRITE PSEUDOCODE → TRACE → REFINE → IMPLEMENT.

ANSWERS – MOBILE COMPUTING ASSIGNMENT 3 (CO3)

Values below are worked analytical examples with stated assumptions.

Q1 – Wireless Network Architecture and Mobility Analysis

Part A – Shared work and Student A (architecture)



- **Components:** mobile nodes (move and communicate), AP/base station (radio access, channel control, bridge to wired network), wireless links (shared, limited bandwidth, error-prone), gateway/router (to backbone), server/CN.
- **Characteristics:** shared broadcast medium, high error rate, variable bandwidth, limited range, mobility, battery limits.
- **Bandwidth effect:** cell capacity C is shared, so each of n users gets about C/n (10 Mbps and 10 users: 1 Mbps). More load gives queuing delay and loss. Weak signal lowers rate and causes disconnection.
- **Flow:** MN → AP → gateway → Internet → CN, and back. MN reaches the network only through the AP.

Student B – Mobility and performance (cell diameter 400 m, handoff gap 0.3 s, 100 packets/s)

Scenario	Speed	Dwell time	Handoffs/min	Loss/min	Availability
Static	0	–	0	0%	100%
Pedestrian	1.4 m/s	286 s	0.21	0.1%	99.9%
Vehicle	15 m/s	26.7 s	2.25	1.1%	98.9%
Fast vehicle	30 m/s	13.3 s	4.5	2.3%	97.7%

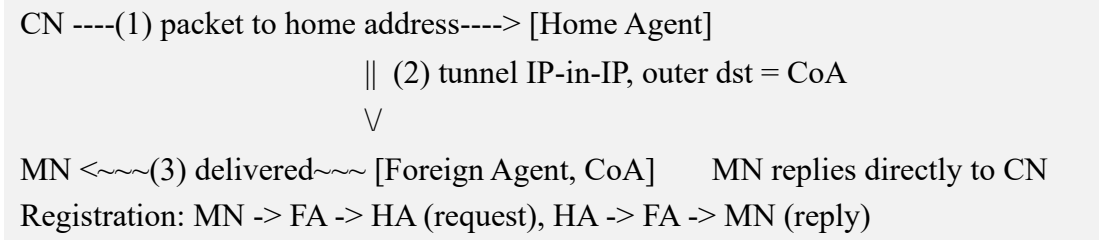
Dwell time = diameter / speed. Loss per handoff = 100 × 0.3 = 30 packets. Higher speed means more handoffs, more loss and delay; voice suffers more than file transfer.

Part B – Mobility scenario (user moves from AP1 to AP2)

- **Student A:** successful = signal above threshold, packets acknowledged, or handoff done in the overlap. Interrupted = left AP1 before attaching to AP2, or association fails. **Events:** attached to AP1 → RSSI below -80 dBm → scan → find AP2 → associate → update address/routing → resume. Delay about 20

  ms normal, about 300 ms in handoff; 30 packets lost. Delivery 99.9%, 98.9%, 97.7% as speed rises.
- **Student B:** handoff when RSSI(current) < -80 dBm AND RSSI(candidate) > RSSI(current) + 5 dB hysteresis (avoids ping-pong). Affected: delay, loss, throughput, IP binding. Scenario ends on HANDOFF_SUCCESS, on DISCONNECTED (no AP) or session end. Efficiency = 1 - handoffs/s × handoff time = 1 - 0.0225 = 97.7% at 4.5 handoffs/min.

Q2 – Mobile IP and Mobility Management



- **MN:** keeps home address, detects movement, gets CoA, registers. **HA:** keeps binding (home address → CoA), authenticates, intercepts and tunnels packets. **FA:** advertises, gives CoA, relays registration, decapsulates and delivers. **CN:** sends to home address (unchanged; with route optimisation it uses the CoA).

- **Agent discovery:** HA/FA send periodic Agent Advertisements; MN may send a Solicitation. **Registration:** Request (home address, CoA, lifetime, identification) → HA authenticates, updates binding → Reply. **Delivery:** CN → home network → HA encapsulates (20-byte outer header) → FA decapsulates → MN.
- **Worked example:** MN 10.1.1.5 moves to network 20.2.2.0/24, FA/CoA 20.2.2.1, HA 10.1.1.1. Binding 10.1.1.5 → 20.2.2.1. Packet from CN 30.3.3.3 to 10.1.1.5 becomes outer 10.1.1.1 → 20.2.2.1, then the FA delivers the inner packet to the MN.

Student B – Challenges

Challenge	Analysis
Registration delay	2×5 + 2×40 + 10 = about 100 ms (more if HA is far); packets go to the old CoA meanwhile
Tunnelling overhead	20 B per packet: 20% of a 100 B packet, 2% of 1000 B
Triangular routing	Via HA 80 ms against 30 ms direct; wastes bandwidth
Packet loss	100 pkt/s × 0.4 s = 40 packets per move
Security / scalability	Authenticate registration (HMAC), replay protection (identification), IPsec tunnel; HA is a bottleneck and single point of failure

Joint design question – frequent change of network during a session

Problems: repeated registration signalling and delay, packet loss at each move, tunnelling and triangular-route delay, HA overload, security processing. **Reductions:** route optimisation (CN uses CoA directly); regional/hierarchical registration (local registration about 10–20 ms instead of 100 ms); fast handoff with buffering and forwarding from the old FA; reverse tunnelling where ingress filtering applies; multiple HAs and IPsec. Each fix removes one measured cost above.

Cross-understanding: A explains the challenges and fixes; B explains the full flow: discovery → CoA → registration → binding → HA tunnel → FA delivery → direct reply.

Q3 – Cellular Networks versus Ad Hoc Wireless Networks

CELLULAR: MN ~~ [BS] -- [MSC/Core] -- [BS] ~~ MN AD HOC: N1 -- N2 -- N3 -- N4

(all traffic via base stations) (nodes are hosts and routers, multi-hop)

- **Student A (cellular):** BS gives radio link and channel control; MSC and registers do switching and location. Centralised communication. Handoff: when the serving BS signal drops and a neighbour is stronger, the network moves the call. Frequency reuse and cell splitting give scalability and wide coverage. Uses: voice, mobile broadband, SMS.
- **Student B (ad hoc):** self-organised nodes, no fixed infrastructure. Nodes talk directly or via relays (S → N2 → N4 → D). Topology changes as nodes move, so routes are rebuilt continuously. Uses: disaster rescue, military, temporary and vehicular networks.

Parameter	**Cellular Network**	**Ad Hoc Network**
Infrastructure	Fixed base stations and core	None
Communication	Centralised via BS	Peer-to-peer, multi-hop
Routing	By infrastructure	Distributed (DSDV, AODV)
Mobility	Network-managed	Node route discovery/maintenance
Topology	Planned, static cells	Dynamic
Scalability	High	Limited by overhead and hops
Handoff	Network-controlled between cells	None; routes rebuilt
Parameter	**Cellular Network**	**Ad Hoc Network**
Failure Handling	BS/core failure affects area	Alternate routes via other nodes
Applications	Public voice/data service	Emergency, military, temporary

Decision: ad hoc network for disaster areas (optionally with a gateway to cellular/satellite): no infrastructure needed, quick deployment, multi-hop coverage, no single point of failure. **Cross-defence:** A – cellular needs BS and core for radio resources, switching and handoff, so a BS failure cuts service. B – every node forwards packets and, on failure, sends a route error and finds another path.

Q4 – Applications of Mobile IP and Routing Requirements

Application: mobile healthcare / ambulance. The ambulance (MN) sends vital signs and voice to the hospital server (CN) while moving between networks. Hospital network is home network with the HA; operator networks/hotspots have FAs.

```
Ambulance MN ~~ [Cell/Wi-Fi FA] -- Internet -- [Home Agent] -- [Hospital server CN]
Vitals: MN -> FA -> Internet -> CN.  Instructions: CN -> HA -> tunnel -> FA -> MN
```

- **Student B (architecture):** MN with Mobile IP client, FAs, HA, AAA server, gateways, hospital server, VPN/firewall. Mobile IP keeps the home address fixed, so the session survives network changes. Flow: register CoA → send vitals → doctor replies to home address → HA tunnels to CoA → on new network: discovery, new CoA, re-registration.
- **Student A (performance):** delay direct 40 ms; via HA about 72 ms (within 150 ms voice limit). Interruption per move 0.2–0.4 s; loss = 50 × 0.4 = 20 packets, so 0.7% at one move/min and 2% at three. Requirements: fast movement detection, fast registration, overlapping coverage; security (HMAC, IPsec); reliability (redundant HA, two interfaces, buffering).

Routing requirements: fast binding update to HA, valid route to the CoA after each move, avoid long detours, alternate routes on failure. **Suitability:** Mobile IP is suitable for session continuity but needs fast handoff, route optimisation and security for critical use.

Q5 – Proactive Routing Protocols in Mobile Networks

Proactive routing keeps routes to all destinations through periodic and triggered updates. **DSDV** uses (destination, next hop, metric, sequence number); a route is accepted if its sequence number is higher, or equal with a smaller metric. Destinations issue even sequence numbers; a broken link gets metric infinity and an odd number. Topology: chain A–B–C–D–E.

Table of B	**Before**	**After C–D breaks**	**After D reconnects to B**
A	A, 1, A-100	A, 1, A-100	A, 1, A-100
C	C, 1, C-100	C, 1, C-100	C, 1, C-100
D	C, 2, D-100	– , ∞, D-101	D, 1, D-102
E	C, 3, E-100	– , ∞, E-101	D, 2, E-102

(next hop, metric, sequence number)

```
ALGORITHM UpdateDSDVRoute(Node, ReceivedRoute)
BEGIN
        NewHop ← ReceivedRoute.HopCount + 1 ; NewSeq ← ReceivedRoute.SequenceNumber
 IF Destination not in RouteTable THEN ADD ROUTE
 ELSE IF NewSeq > RouteTable[Destination].SequenceNumber THEN UPDATE ROUTE
 ELSE IF NewSeq = RouteTable[Destination].SequenceNumber AND NewHop < RouteTable[Destination].HopCount THEN UPDATE ROUTE
 END IF
 RETURN RouteTable
END
```

- **Student B:** overhead – each node sends N entries per period, so about N×N in the network (N = 10, 50, 100 gives 100, 2,500, 10,000 entries). Route availability is high, but stale after a break until updates arrive. Convergence about diameter × update interval. Delay is low (no discovery); delivery falls at high mobility because of stale routes and control traffic.

Q6 – Reactive Routing and Protocol Comparison

- **Student A (AODV):** routes are found only on demand. **RREQ:** source floods (source, destination, request ID, sequence numbers); nodes record reverse routes and discard duplicates. **RREP:** destination (or node with fresh route) replies along the reverse path, installing forward routes. **RERR:** sent to upstream nodes when a link fails. Maintenance: monitor links, local repair or RERR, rediscover if data remains; unused routes expire. Mobility causes more RERR and rediscovery on active routes only.

Step	Node	Event	Route state
1	S	Needs D, no route; broadcasts RREQ	DISCOVERY
2	A	Records reverse route, forwards	DISCOVERY
3	D	Receives RREQ, sends RREP	VALID
4	S	Installs S→A→D, sends data	ACTIVE
5	A	Link A–D fails, sends RERR	INVALID → rediscovery

Aspect	DSDV	AODV
Discovery	None; table updates	RREQ/RREP on demand
Routing table	All destinations	Only used routes
Overhead	Continuous, about $O(N^2)$	On demand; floods per discovery
Initial delay	Very low	Higher (one round trip)
Mobility	Updates network-wide; stale routes	Repairs only affected routes
Suitable for	Small, stable, heavy traffic	Dynamic, larger, bursty traffic

Joint decision: AODV. DSDV overhead grows with N^2 and with every topology change, and its tables go stale; AODV generates control traffic only for active or broken routes and gives fresh routes. The cost is a discovery delay, which is accepted for better delivery.

Q7 – Design a Mobile Emergency Communication Network

Stage 1 – Independent proposals (retained)

- **Student A:** hybrid network – ad hoc Wi-Fi/radio mesh of vehicles and teams, with gateway vehicles/drones linked to the command centre by cellular/satellite. Mobility: permanent home addresses, Mobile IP with FAs on gateways, regional registration. Mobile IP keeps responders reachable with the same address. Model: client–server to command centre plus peer-to-peer group voice.
- **Student B:** reactive AODV (changing topology, bursty traffic). Maintenance: hello messages, RERR, local repair, route timers. Failure handling: alternate route or rediscovery; buffering and retry on partition; drones as relays; priority

  for critical messages. Parameters: delay, packet delivery ratio, routing overhead, discovery latency, throughput, energy.

Stage 2 – Comparison

Agree: no dependence on fixed infrastructure; nodes relay. **Differ:** A handles location/addressing, B handles path finding, so they are complementary. **Strengths:** A gives session continuity; B adapts quickly with low idle overhead. **Weaknesses:** A

needs reachable HA/FA, registration delay, tunnel overhead; B has discovery delay and flooding overhead. Mobile IP needs agent advertisements carried over AODV. Expected overhead: registration + 20 B tunnel header + RREQ floods. Expected delay: about 100 ms registration, 50–300 ms first-packet discovery. Mobility: re-registration plus RERR/rediscovery.

Stage 3 – Joint final design

```
MN1 Ambulance ~~~\
      MN2 Police ~~~~~~~> Relay nodes (AODV) ~~~> Gateway/FA
---(cellular/satellite)--> HA + CN
MN3 Rescue team ~~/                (vehicle/drone)          (Command centre)
```

- **Mobile IP:** MN = responder, HA and CN at command centre, FA = gateway. FA advertisements flood the ad hoc network; MN registers through its AODV route to FA, FA relays to HA; HA tunnels (IP-in-IP) to CoA; FA decapsulates and forwards by AODV; local re-registration inside an area.
- **Routing:** AODV (dynamic topology, bursty traffic, low state). **Failure handling:** link/node failure → RERR, local repair or rediscovery; gateway failure → new FA, re-register; disconnection → buffer and retry, backup HA and gateways.

Stage 4 – Validation

- **Student A (200 B packets, 50 pkt/s, registration 100 ms, handoff 300 ms, 3 hops × 10 ms):** tunnel overhead 20/200 = 10%. Delay 30 ms normal, 72 ms via HA, 400 ms once per move. Loss 50 × 0.4 = 20 packets per move; one move per 30 s gives PDR 98.7%, above 99.5% with buffering. Interruption 0.4 s, 0.32 s with regional registration, under 0.1 s with fast handoff. Mobile IP with regional registration keeps address and session with low overhead.
- **Student B (test design):** nodes 10, 25, 50, 100; speeds 1, 5, 20 m/s with pause times 0, 30, 60 s; failure tests (gateway, node, partition); run AODV and DSDV, 300 s, 5 runs, record delay, PDR and overhead (fill from your simulator output).

Condition	**AODV (expected)**	**DSDV (expected)**
Low mobility	PDR above 95%, low overhead	PDR high, lowest delay
Condition	**AODV (expected)**	**DSDV (expected)**
High mobility	PDR about 85–95%	PDR falls, overhead rises
Many nodes	Moderate overhead growth	Overhead grows about N^2
Failure	Recovers in under 1 s	Recovers after update spreads

Stage 5 – Cross-defence

A – why AODV: topology changes constantly, so proactive tables would be stale and waste bandwidth and battery; AODV finds fresh routes on demand, repairs only broken routes and needs no infrastructure. **B – how Mobile IP maintains communication:** the MN keeps its home address; after a move it gets a new CoA and registers it with the HA; the HA tunnels traffic for the home address to the new CoA; only the short registration time is lost.

Stage 6 – Joint evaluation

- **Architecture suitable:** no fixed infrastructure, quick set-up, multi-hop coverage, no single point of failure, gateways to command centre. **Mobile IP required** for stable addressing and sessions across areas; not needed if all traffic stays inside one ad hoc network. **AODV appropriate** (Stage 5).
- **Overhead:** 20 B per tunnelled packet (10%), registrations, RREQ floods and hellos. **Mobility effect:** about 0.3–0.4 s gap per move, route breaks add delay and loss. **Connectivity lost:** buffer and retry, RERR and rediscovery, switch gateway, local communication continues in each partition.
- **Not appropriate when:** very large dense networks, routes break faster than discovery, hard real-time limits below discovery/handoff delay, no gateway or HA reachable, or heavy steady traffic to many destinations (proactive better).

Conclusion: the designer should analyse infrastructure availability, mobility speed and handoff frequency, network size, traffic type and rate, delay and loss tolerance, bandwidth and battery, need for session continuity, overhead tolerance, security, and failure/partition behaviour. Frequent mobility with bursty traffic favours Mobile IP with regional registration and AODV; stable networks with heavy steady traffic favour proactive routing.

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